SafeCity Dashboard — Production Server Specs

Target: ~3000+ concurrent users | Mixed read/write workload | Large-scale uploads (TB growth expected)

SECTION	DETAILS
Architecture	Load Balancer → 4–6 Django App Nodes → PostgreSQL (PostGIS) → Object Storage (CDN)
Backend Stack	Django + DRF + JWT	GeoDjango (PostGIS)	REST APIs (~53 groups)
Frontend	React (Vite) deployed via CDN/static hosting
App Servers	4–6 nodes	4 vCPU	8–16 GB RAM	Stateless	No media storage
Database	PostgreSQL + PostGIS	16–32 vCPU	64–128 GB RAM	2–4 TB NVMe SSD	HA + Backups
Storage	S3-compatible object storage	Multi-TB	CDN for fast delivery
Networking	Public: Load Balancer only	Private: App + DB	TLS enforced
Performance Targets	APIs: <300 ms avg	Heavy endpoints (Gantt): optimized + cached
Scaling Strategy	Horizontal scaling (add app nodes)	DB tuning + read replicas if needed
Critical Optimizations	Redis caching	Pagination	Async tasks (Celery)	Query optimization
Media Handling	Do NOT use Django for media in production	Use Object Storage or Nginx
Security	DEBUG=False	Secure env vars	Restricted DB access	HTTPS only
GPU Requirement	NOT required. Only needed for AI/ML workloads (e.g., image processing, ML models, video analytics).

Note: Avoid single-server deployment at this scale. Use distributed architecture with monitoring and backups.